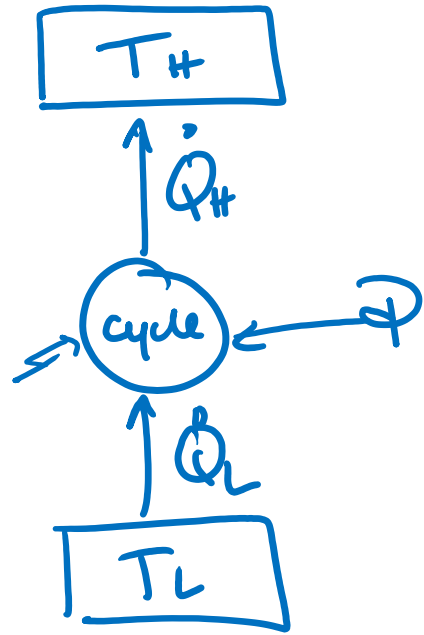


ME 300:
Engineering Thermodynamics

Vapor
Compression Cycle

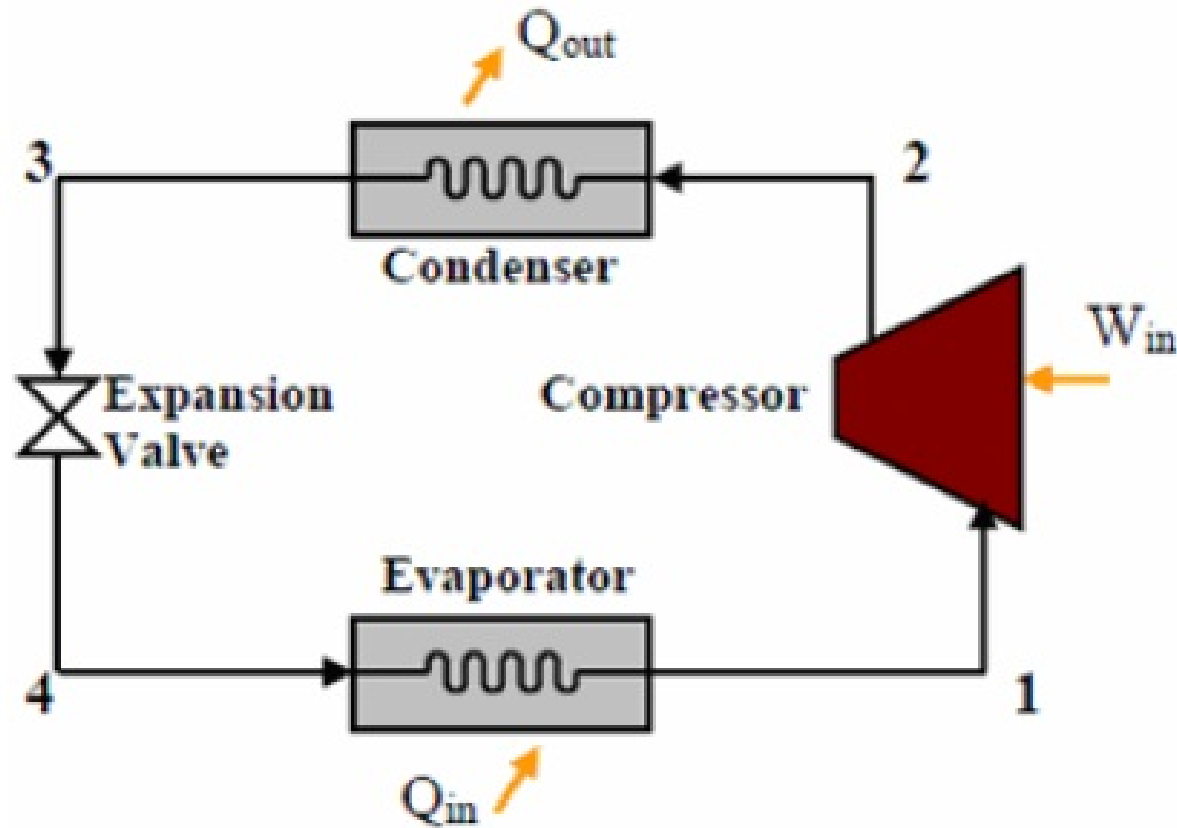


Vapor compression



refrigeration: $\beta = \frac{\dot{Q}_L}{\Phi}$

Vapor compression *-ideal*



Vapor Compression Cycle

1-2: isentropic compression
adiabatic, reversible

=> compressor, vapor

2-3: isobaric heat rejection

=> condenser, vap → liq

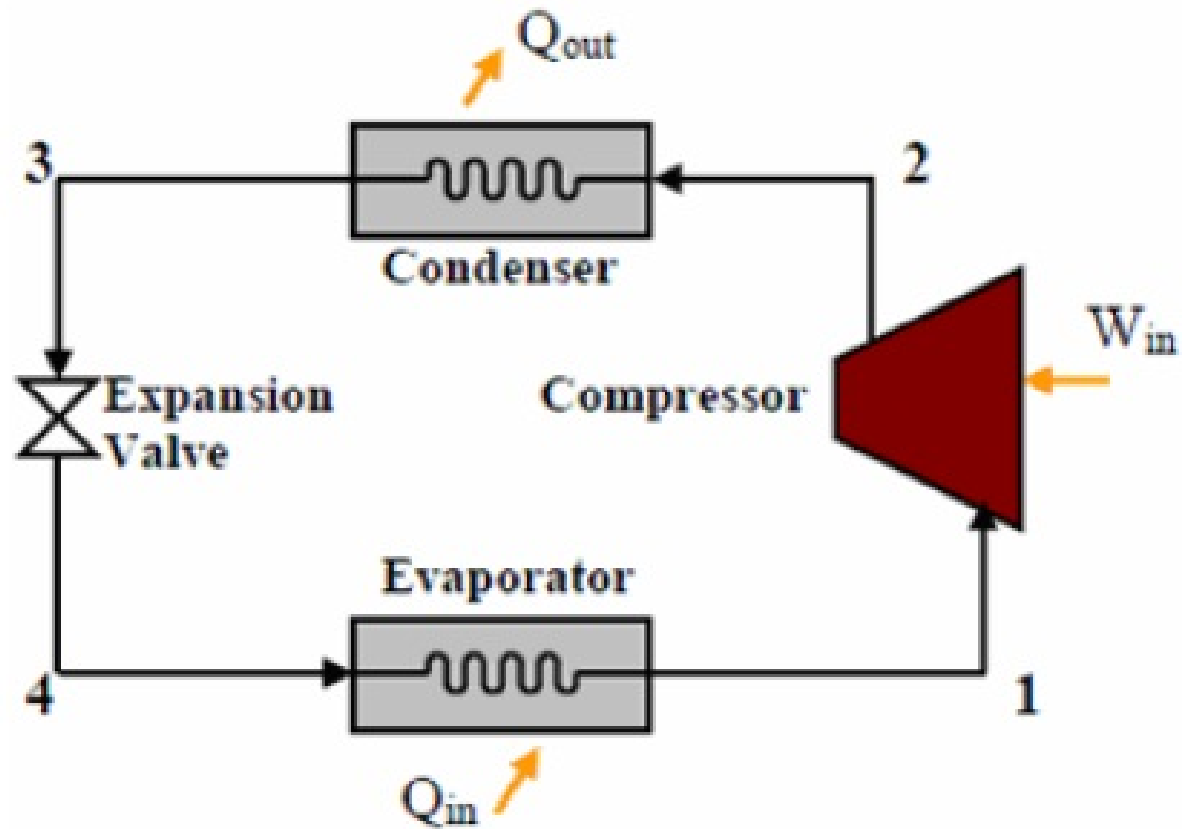
3-4: valve expansion, $h_3 = h_4$

=> valve, liq → mixture

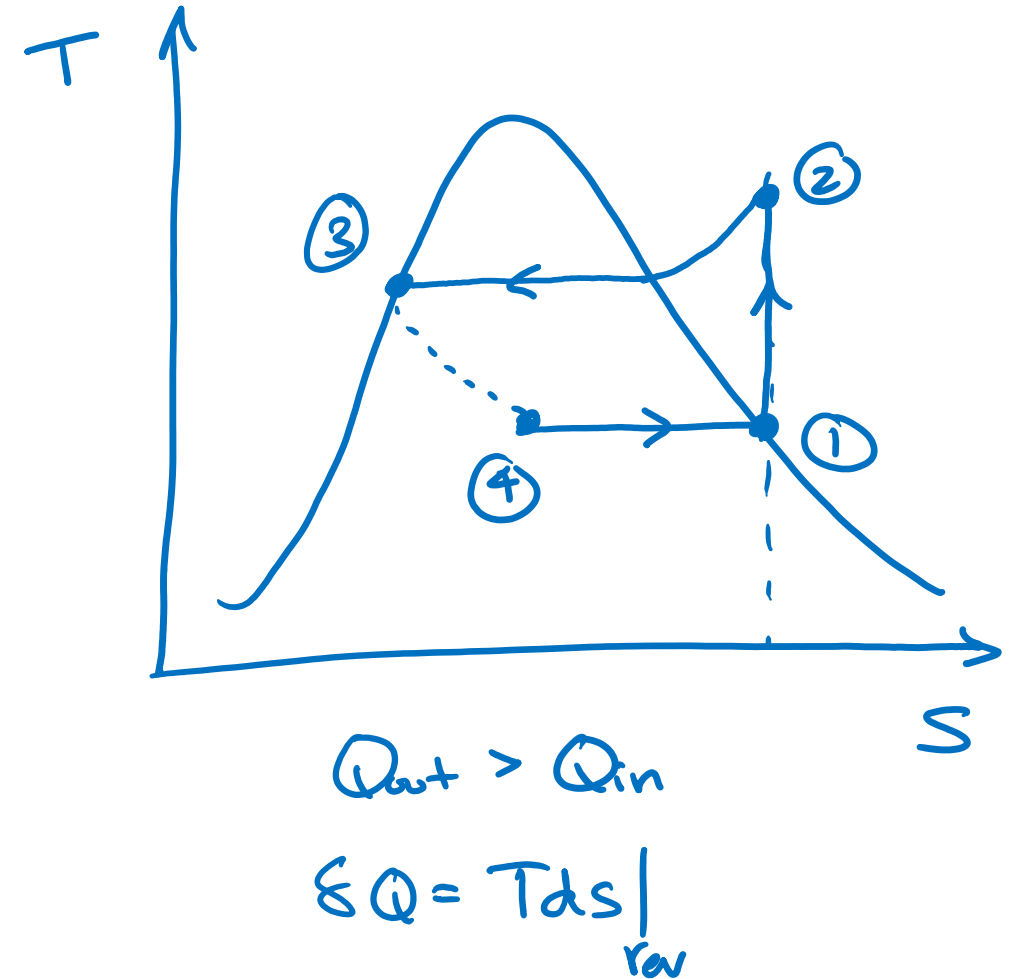
4-1: isobaric heat addition

=> evaporator → mix → vap

Vapor compression



Vapor Compression Cycle



Energy conservation steady-flow $\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$

$$1-2: \text{compressor} \Rightarrow \dot{m}(h_2 - h_1) = -\dot{W}_2$$

$$2-3: \text{condenser} \Rightarrow \dot{m}(h_3 - h_2) = {}_2\dot{Q}_3$$

$$3-4: \text{throttle} \Rightarrow h_4 = h_3$$

$$4-1: \text{evaporator} \Rightarrow \dot{m}(h_1 - h_4) = {}_4\dot{Q}_1$$

$$\beta = \frac{{}_4\dot{Q}_1}{\dot{Q}} = \frac{{}_4\dot{Q}_1}{|\dot{W}_2|}$$